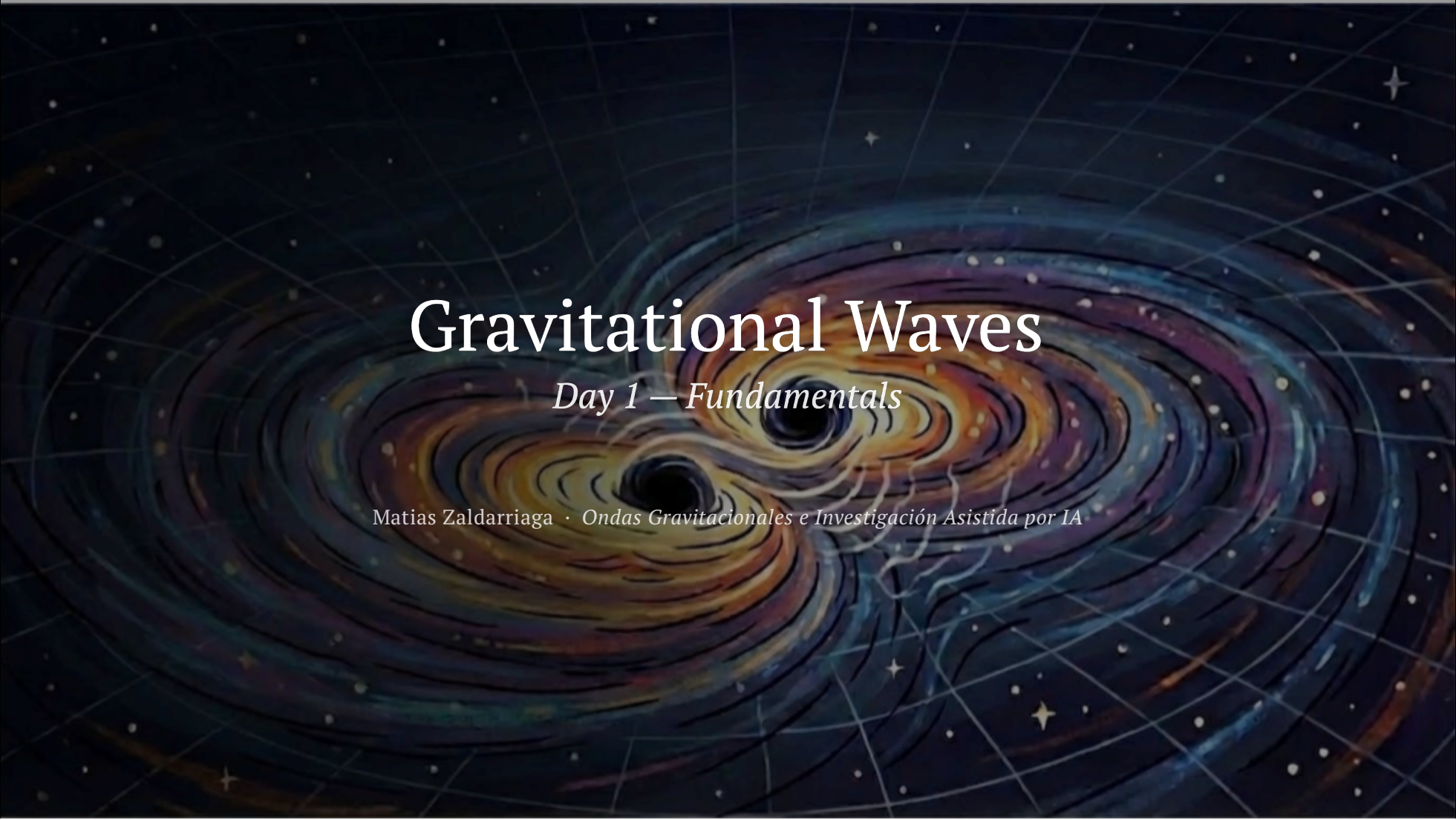


The background of the slide is a dark blue space filled with a grid of white lines representing spacetime. Two large, swirling, colorful ripples (representing gravitational waves) are centered in the image, with colors ranging from dark blue to yellow and orange. Numerous small white stars are scattered throughout the background.

Gravitational Waves

Day 1 — Fundamentals

Matias Zaldarriaga · *Ondas Gravitacionales e Investigación Asistida por IA*

What this week is.

THE SCIENCE

- **Day 1** — Fundamentals. Where the waves come from.
- **Day 2** — Ground-based detection. How we hear them.
- **Day 3** — From observations to astrophysics.
- **Day 4** — Pulsar timing arrays. The nanohertz sky.
- **Day 5** — Open problems.

THE WAY OF WORKING

- **Day 1** — What an agent *is*.
- **Day 2** — Navigating code nobody in the room knows.
- **Day 3** — Scaling out: jobs, remote, parallel.
- **Day 4** — More than one of them. Roles, and an independent check.
- **Day 5** — Turning a result into an output.

*Each day's practical half is built around **papers**. We read them, and then we make an agent reproduce them — which is a far more demanding test of understanding than reading.*

This is **not** a course on the right way to use AI. Nobody knows the right way; the field moves faster than any syllabus. It is a catalogue of things one person actually does, offered as an entry point.

In two halves.

THIS SESSION

Where gravitational waves come from, what they do, how loud they are, and how far the quadrupole formula carries you across ten decades of mass.

Almost everything will be an estimate. Factors of two are beneath us today.

THE SESSION AFTER IT

You point an agent at a real paper and try to reproduce it. Then we compare what everyone got.

Bring the laptop you ran `doctor.py` on.

The two papers you will meet next are Peters (1964) and The basic physics of the binary black hole merger GW150914 (2016). Everything in this lecture is chosen to make those two papers reproducible.



SECTION I

Spacetime is dynamical

Gravity is *geometry*.

Newton: gravity is a force acting instantaneously at a distance. Einstein: there is no force. Free particles move on straight lines — geodesics — in a spacetime whose geometry is itself determined by the matter in it.

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

The left-hand side is built from second derivatives of the metric $g_{\mu\nu}$. That is the crucial structural fact: **the field equations are second order in time**. A theory whose fundamental field obeys a second-order equation of motion has *waves*.

Newtonian gravity has no waves because it has no dynamics — $\nabla^2\Phi = 4\pi G\rho$ is an instantaneous constraint. The moment you make gravity consistent with a finite speed of propagation, you are forced into a wave equation.

Linearise.

Far from any source, spacetime is nearly flat. Write

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1$$

and keep only terms linear in h . In a convenient gauge, Einstein's equations collapse to the flat-space wave equation:

$$\square \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu} \quad \Longrightarrow \quad \square \bar{h}_{\mu\nu} = 0 \text{ in vacuum}$$

Ripples in the geometry, propagating at c . Two independent polarisations survive after all the gauge freedom is used up — the *transverse-traceless* components.

Ten components in $h_{\mu\nu}$; four coordinate freedoms, four residual gauge freedoms — two left. A massless spin-2 field has two helicity states, exactly as a photon has two.

The wave, written down.

For a wave travelling along $\hat{z}$:

$$h_{\mu\nu}(t, z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \cos(\omega t - kz)$$

Transverse — nothing in the z row or column. **Traceless** — the xx and yy entries are equal and opposite. Two amplitudes, h_+ and $h_\times$, rotated 45° from one another. They are called the **strain**, and they are dimensionless.

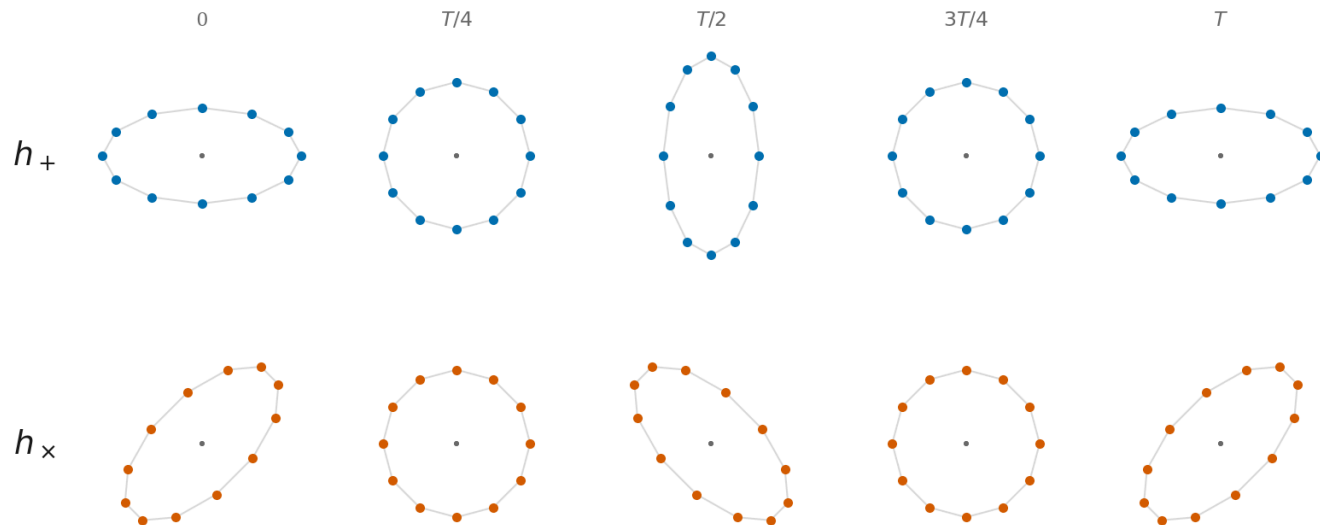
Electromagnetism

Spin 1. Polarisation basis rotates back into itself under $360^\circ/1$.
Dipole radiation dominates.

Gravity

Spin 2. Rotate by 180° and you are back where you started.
Quadrupole radiation dominates.

What a wave *does*.



A ring of free test masses under a passing wave, over one period.

A single free particle does nothing interesting — you can always choose coordinates in which it stays put.

Gravity is detectable only as a **tidal** effect: the change in the separation of *two* free particles, $\Delta L/L = \frac{1}{2}h$.

Squeezed along one axis, stretched along the other, then the reverse. The two polarisations are the same pattern rotated by 45° .

How small is h ?

For the events LIGO detects, $h \sim 10^{-21}$. Over a 4 km arm:

$$\Delta L \sim h L \sim 10^{-18} \text{ m} \left(\frac{L}{\text{km}} \right) \left(\frac{h}{10^{-21}} \right)$$

That is $\sim 10^{-3}$ of the diameter of a proton. Some comparisons worth carrying around:

Quantity	Precision achieved
Electron magnetic moment	10^{-13}
Best atomic clocks (fractional frequency)	10^{-19}
LIGO strain	10^{-21}

And yet the *phase* shift on the laser is what is actually measured —

$$\frac{\Delta L}{\lambda_{\text{laser}}} \sim 10^{-12} \left(\frac{L}{\text{km}} \right) \left(\frac{h}{10^{-21}} \right) \left(\frac{1064 \text{ nm}}{\lambda_{\text{laser}}} \right)$$

The Fabry–Pérot cavities buy a factor of ~ 250 in effective arm length. The rest is photon statistics: with $\dot{N} \sim 4 \times 10^{21}$ photons per second, $\sqrt{N}$ is a large number. (All of this is day 2.)

Weak coupling is a *feature*.

The same weakness that makes gravitational waves nearly impossible to detect makes them uniquely valuable: **the Universe is transparent to them.**

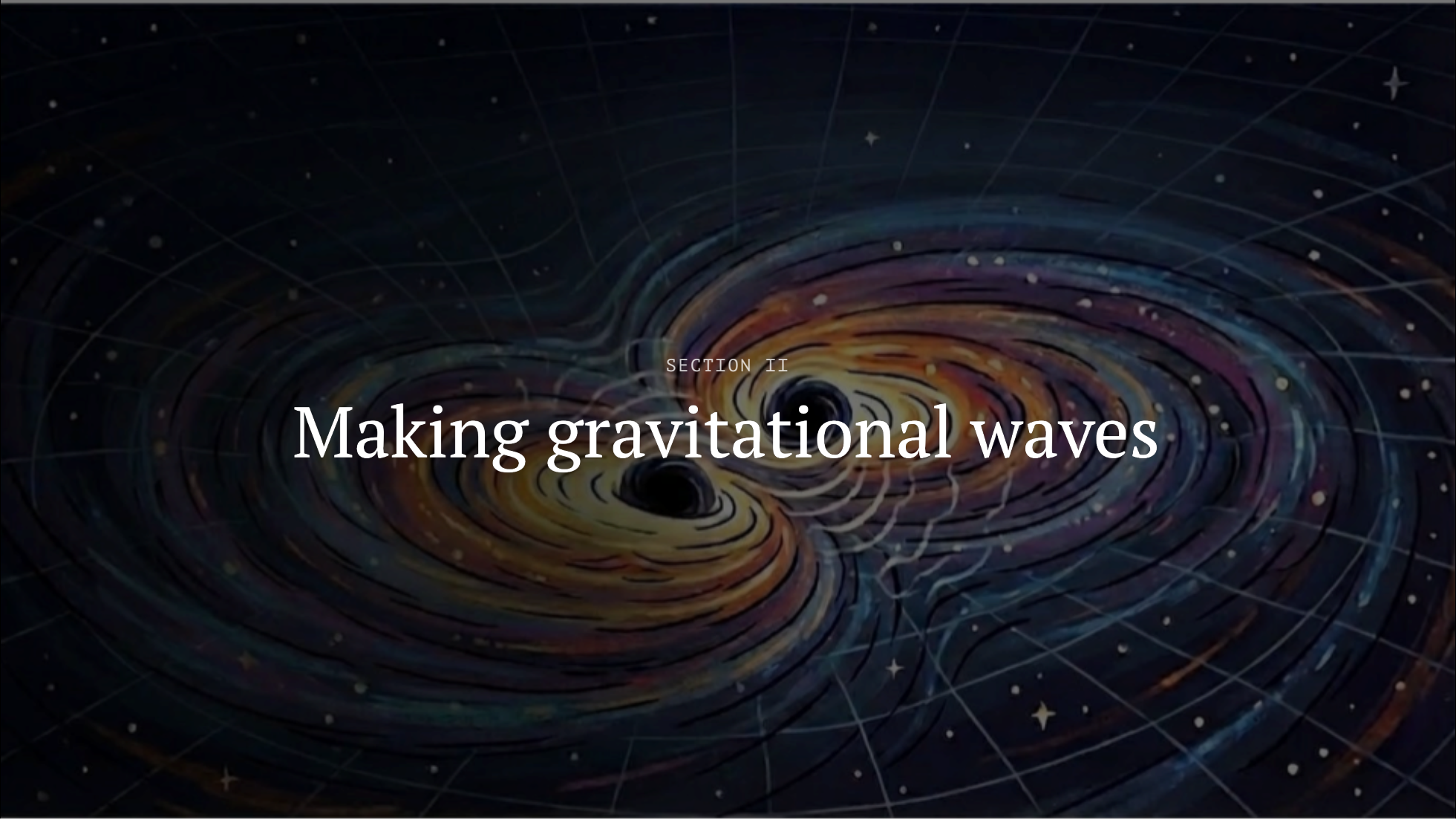
PHOTONS

Scatter, absorb, get reprocessed. Before recombination the Universe is opaque. We see the surface of things.

GRAVITATIONAL WAVES

Propagate essentially unimpeded from the source to us. We see the *dynamics* of the mass itself, from inside the horizon of a merging black hole to the first second of the Universe.

A gravitational-wave detector is not a telescope. It has almost no angular resolution and it is sensitive to the whole sky at once. It is much closer to a microphone.

A visualization of gravitational waves as ripples in spacetime. The background is a dark blue space filled with a grid of lines representing spacetime. Two large, swirling, colorful ripples (representing gravitational waves) are shown, with colors ranging from dark blue to yellow and orange. The ripples are centered around two black holes that are in the process of merging. The text "SECTION II" is centered above the main title.

SECTION II

Making gravitational waves

Why *quadrupole*?

In electromagnetism the leading radiation is **dipole**, $\ddot{\vec{d}}$ with $\vec{d} = \sum q_i \vec{r}_i$. In gravity the analogous quantities are conserved:

Mass monopole

$\sum m_i$ = total mass. Conserved $\Rightarrow$ no monopole radiation.

Mass dipole

$\sum m_i \vec{r}_i$ = centre of mass. Its second derivative is $\dot{\vec{P}} = 0 \Rightarrow$ no dipole radiation.

The first non-vanishing term is the **quadrupole**. This is not an accident of the equations: it is momentum conservation, and it is the field-theory statement that the graviton has spin 2.

$$h_{ij}^{\text{TT}} \simeq \frac{2G}{c^4 d} \ddot{Q}_{ij}, \quad Q_{ij} = \int \rho \left(x_i x_j - \frac{1}{3} \delta_{ij} r^2 \right) d^3x$$

*Consequence: a spherically symmetric collapse radiates **nothing**, however violent. A supernova is a poor gravitational-wave source unless it is asymmetric.*

Units: set $G = c = 1$.

Every estimate today is easier in geometrised units. Masses become lengths, velocities become dimensionless:

$$r_s = \frac{2GM}{c^2} = 2M$$

Object	Mass	As a length	As a time
Sun	$1 M_\odot$	1.5 km	5 μ s
GW150914	$65 M_\odot$	96 km	0.32 ms
Sgr A*	$4 \times 10^6 M_\odot$	0.04 AU	20 s
M87*	$6.5 \times 10^9 M_\odot$	64 AU	9 hr

Read the last column as "*the shortest timescale on which this object can do anything.*" It is already, to a factor of a few, the gravitational-wave period at merger.

The Newtonian *binary*.

DEFINITIONS

$$M_T = m_1 + m_2, \quad \eta = \frac{m_1 m_2}{M_T^2}, \quad \mu = \eta M_T$$

$\eta \leq 1/4$, with equality for equal masses. This bound does more work than you would expect.

CIRCULAR ORBIT, SEPARATION A

$$\Omega^2 = \frac{M_T}{a^3}, \quad v = (M_T \Omega)^{1/3}, \quad |E| = \frac{1}{2} M_T \eta v^2$$

Kepler, in geometrised units. v is the relative orbital velocity.

The gravitational-wave frequency is **twice** the orbital frequency — the mass distribution of a binary returns to itself after half a turn:

$$f_{\text{GW}} = 2 \frac{\Omega}{2\pi}$$

That factor of two is spin-2 again, and it is one of the most common places to lose a factor of 2. Watch for it later.

The amplitude, and the *chirp mass*.

Put the binary into the quadrupole formula. $Q \sim \mu a^2$, and each time derivative brings down Ω :

$$h \sim \frac{\ddot{Q}}{d} \sim \frac{M_T \eta v^2}{d} \sim \frac{M_T \eta (\Omega M_T)^{2/3}}{d} \sim \frac{\mathcal{M}_c^{5/3} \Omega^{2/3}}{d}$$

The two masses enter **only** through one combination:

$$\mathcal{M}_c \equiv \eta^{3/5} M_T = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}}$$

the **chirp mass**. It is the best-measured parameter of almost every event, and the reason individual masses are so much harder to pin down than you would guess from how loud the events are.

With all the constant factors: $h = \frac{4}{d} \mathcal{M}_c^{5/3} (\pi f)^{2/3}$, angle-averaged.

The *luminosity*.

The energy flux in a wave goes like $\dot{h}^2$, integrated over a sphere of radius d :

$$P_{\text{GW}} \sim \dot{h}^2 d^2 \sim (\Omega M_T \eta v^2)^2 \sim \eta^2 v^{10}$$

In geometrised units P is **dimensionless** — the mass has cancelled completely. Restoring units, the natural scale is

$$L_{\text{P}} = \frac{c^5}{G} \approx 3.6 \times 10^{52} \text{ W}$$

set by G and c alone. No mass, no length, no size of anything. A merging binary radiates a fixed *fraction* of this, determined only by the dimensionless numbers η and v/c .

For comparison: the entire electromagnetic output of the observable Universe is $\sim 10^{49}$ W. A binary black hole merger briefly outshines it by a factor of a thousand — in gravitational waves.

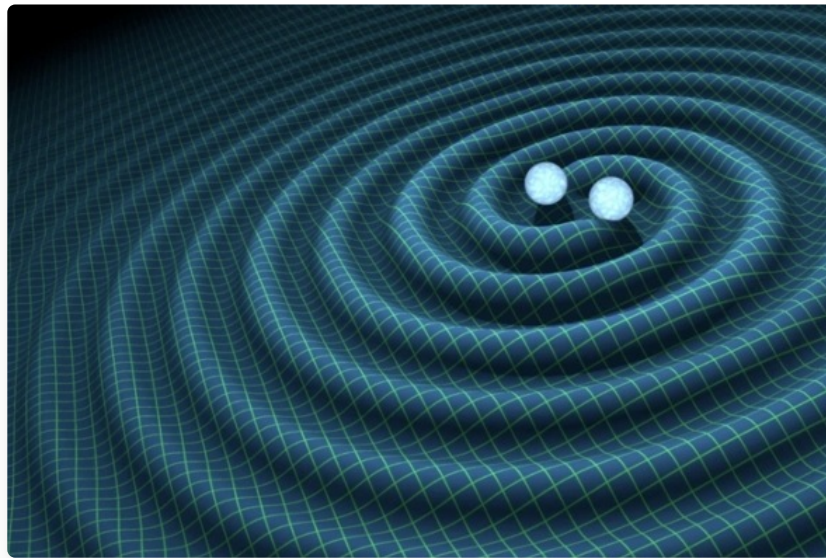
Why *compact* objects.

$P_{\text{GW}} \propto v^{10}$ is a brutal scaling. To radiate anything at all you need v/c of order one, and for a bound orbit

$$\frac{v^2}{c^2} \sim \frac{GM}{c^2 a} \sim \frac{r_s}{a}$$

so $v/c \sim 1$ **means the objects are nearly touching their own Schwarzschild radii**. Only white dwarfs, neutron stars and black holes qualify.

The Earth–Sun system: $v/c \sim 10^{-4}$, so $P_{\text{GW}} \sim 200$ W. It will never be measured, and the orbit decays in $\sim 10^{23}$ years.



Spacetime ripples from an inspiralling binary — T. Pyle / Caltech / MIT / LIGO Lab

The orbit *decays*.

Energy is leaving, so the orbit shrinks. The timescale:

$$t_{\text{GW}} \sim \frac{E}{P_{\text{GW}}} \sim \frac{M_T \eta v^2}{\eta^2 v^{10}} \sim M_T \eta^{-1} v^{-8} \sim \mathcal{M}_c^{-5/3} \Omega^{-8/3}$$

A more useful dimensionless statement — how much does the system change in **one orbit**?

$$\frac{\dot{E}}{E} \times P \sim \frac{P_{\text{GW}}}{\Omega E} \sim \eta v^5 \quad \implies \quad \dot{P} = -2\pi \frac{96}{5} (\mathcal{M}_c \Omega)^{5/3}$$

$\dot{P} \sim 1$ means the period changes by order itself in one period. That is the merger. Everything before it is a slow, adiabatic inspiral — which is why a **post-Newtonian expansion in v/c works at all**.

Peters (1964) is precisely the calculation of $\dot{a}$ and $\dot{e}$ from this argument, done properly, including eccentricity. That is your first paper.

The chirp.

FREQUENCY EVOLUTION

$$\frac{\dot{\Omega}}{\Omega} \sim \frac{P_{\text{GW}}}{E} \sim \frac{\eta v^8}{M_T} \implies \Omega \sim \mathcal{M}_c^{-5/8} \tau^{-3/8}$$

with τ the time to merger. The frequency sweeps upward, ever faster — the *chirp*.

CYCLES LEFT

$$N = \int f d\tau \sim (\mathcal{M}_c f)^{-5/3}$$

GW150914 spent ~ 10 cycles in band. A neutron-star binary spends $\sim 10^4$.

IN THE FREQUENCY DOMAIN

Stationary phase: the Fourier integral $\int h_0 e^{i\Phi(t)} e^{-2\pi i f t} dt$ is dominated by the moment when the instantaneous frequency equals f , and the window around it has width $\sim 1/\sqrt{\dot{f}}$ — that is what the time integration leaves behind. So

$$|h(f)| \sim \frac{h_0}{\sqrt{\dot{f}}} \sim \frac{\mathcal{M}_c^{5/6}}{d} f^{-7/6}$$

Check it by units: $h(t)$ is dimensionless, so $\tilde{h}(f)$ carries seconds — and $h_0/\sqrt{\dot{f}}$ is exactly dimensionless over $\sqrt{\text{Hz s}^{-1}}$, i.e. seconds. With $h_0 \propto f^{2/3}$ and $\dot{f} \propto f^{11/3}$, the exponent is $2/3 - 11/6 = -7/6$. Day 2 plots $\sqrt{f} |\tilde{h}(f)| \propto f^{-2/3}$ against detector noise in the same units.

Later you will read f and $\dot{f}$ straight off a strain plot and get $\mathcal{M}_c$ to 20% with a ruler.

What comes out at *merger*.

ENERGY RADIATED

$$\frac{\Delta M_{\text{rad}}}{M} \sim 3\text{--}10\%$$

Few percent for asymmetric or low spin; up to ~10% for equal-mass, aligned spins.

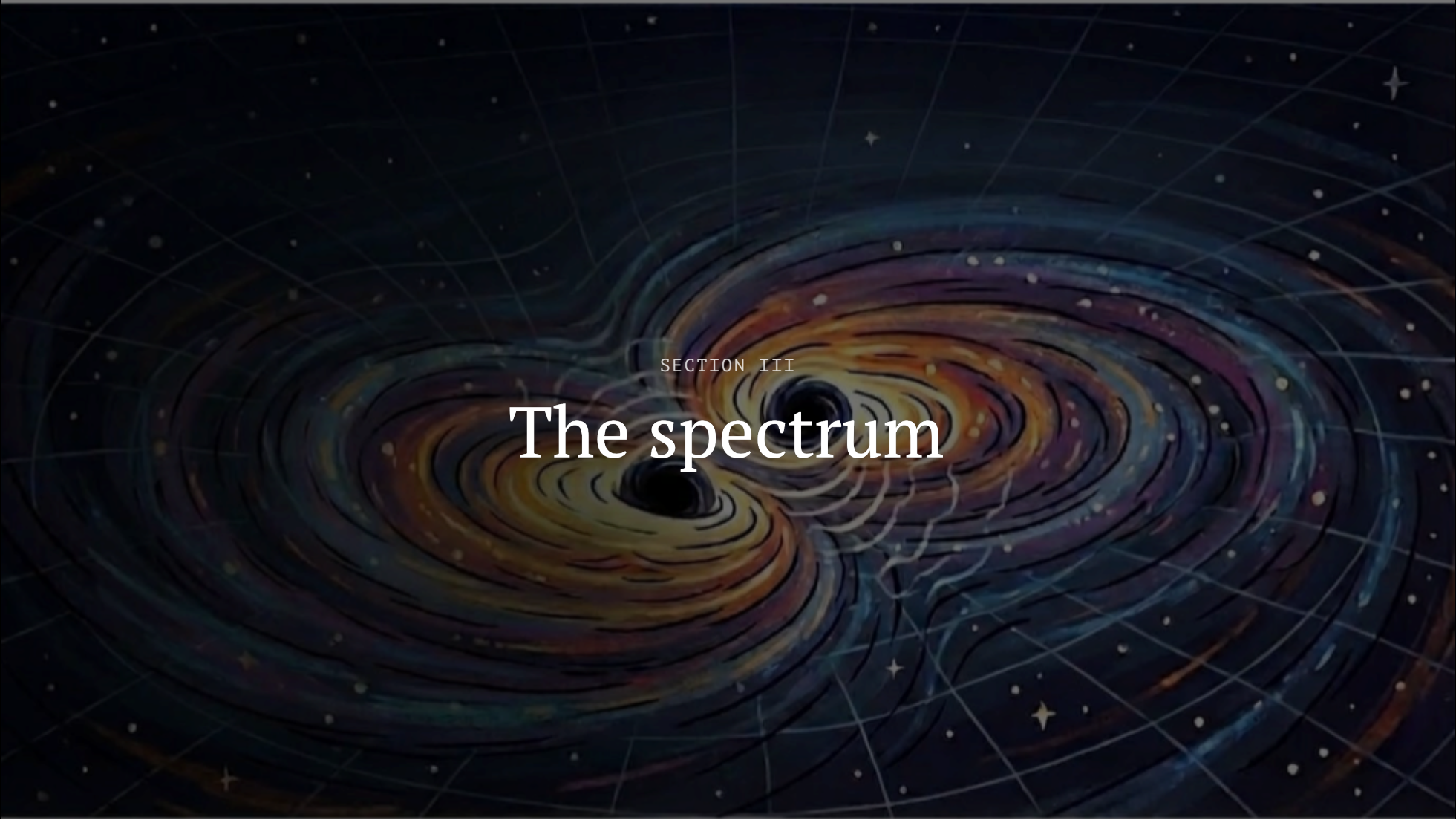
RECOIL OF THE REMNANT

$$\frac{v_{\text{kick}}}{c} \sim 10^{-4}\text{--}10^{-2}$$

Hundreds to a few thousand km/s for precessing-spin "superkicks".

Both are functions of dimensionless numbers only — $q = m_2/m_1$ and the spins χ_1, χ_2 . Not of the total mass. Scale invariance is the organising principle of the whole subject: **a binary black hole merger looks the same at every mass, only faster or slower.**

Which is why the same waveform models serve LIGO at $\sim 10 M_\odot$ and LISA at $\sim 10^6 M_\odot$, and why a PTA source and a LIGO source are the same physics separated by nine orders of magnitude in mass.

The background of the slide is a dark, deep blue space filled with a faint grid of thin, light blue lines. Scattered throughout are numerous small, white, four-pointed stars of varying sizes. In the center, two black holes are depicted as dark, swirling vortices. Each black hole is surrounded by a thick, multi-layered accretion disk. The disks are rendered with vibrant, swirling colors of orange, yellow, and red near the inner edges, transitioning into darker blues and purples as they extend outwards. The two black holes are positioned close together, with their accretion disks overlapping and interacting, creating a complex, dynamic visual effect.

SECTION III

The spectrum

Frequency is set by *mass*.

A binary cannot orbit faster than roughly the light-crossing time of its own Schwarzschild radius. That puts a hard ceiling on the emitted frequency:

$$f_{\text{max}} \sim \frac{c^3}{GM_T} \approx 2 \text{ kHz} \left(\frac{10 M_{\odot}}{M_T} \right)$$

One number determines the band. Read it backwards and a detector's frequency range tells you what masses it can possibly see:

Band	Frequency	M_{max}	What lives there
Audio	10 Hz – 10 kHz	$\sim 10^3 M_{\odot}$	NS and stellar-mass BH mergers
Millihertz	10^{-4} – 1 Hz	$\sim 10^7 M_{\odot}$	massive BH mergers, Galactic white dwarfs, EMRIs
Nanohertz	10^{-9} – 10^{-7} Hz	$\sim 10^{10} M_{\odot}$	supermassive BH binaries

Ten decades in frequency, ten decades in mass, one formula. This is the closest thing gravitational-wave astronomy has to the electromagnetic spectrum — but where the EM spectrum sorts sources by *temperature*, this one sorts them by **mass**.

The *landscape*.

Three experiments, one formula. Sensitivity quoted two ways: the amplitude spectral density $\sqrt{S_n}$ that instrument papers plot, and $h_c = \sqrt{f S_n}$, which is what compares directly against a source.

	Band	Most sensitive at	$\sqrt{S_n}$ there	h_c there	Built to see
PTA (15 yr)	$10^{-9} - 10^{-6}$ Hz	$\sim 1/T \approx 2$ nHz	—	$\sim 2 \times 10^{-15}$	SMBH binaries, $10^8 - 10^{10} M_\odot$
LISA	$10^{-4} - 1$ Hz	~ 7 mHz	1.2×10^{-20}	1.0×10^{-21}	massive BH mergers $10^3 - 10^7 M_\odot$, Galactic white dwarfs, EMRIs
LIGO (design)	10 Hz – few kHz	~ 230 Hz	1.8×10^{-24}	2.7×10^{-23}	NS and stellar-mass BH mergers, $\lesssim 10^3 M_\odot$

Compare against the source table you just saw. GW150914: $h \sim 9 \times 10^{-22}$ against a detector at 2.7×10^{-23} . The PTA source: $h \sim 3 \times 10^{-15}$ against 2×10^{-15} — enormously stronger in absolute terms, and barely detectable.

The reconciliation is h_c . It folds in **how many cycles you get to watch**, and that is where the orders of magnitude are: a LIGO chirp delivers tens of cycles in a fraction of a second, a PTA source delivers a handful over fifteen years. A source can sit far below the noise amplitude and still be found — that is what matched filtering buys, and it is day 2.

Selected *sources*.

Everything on this table follows from three formulas you now have:

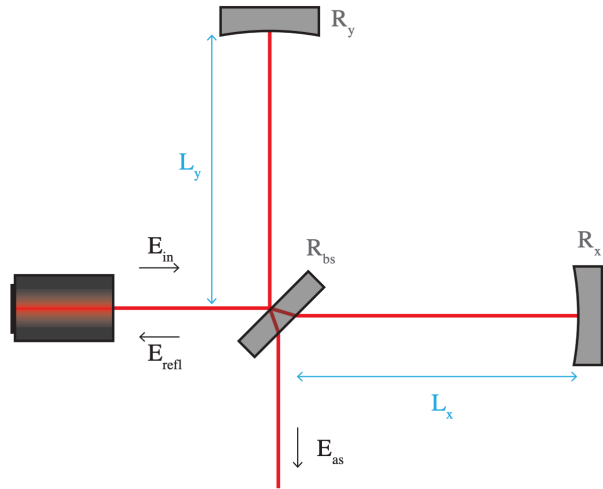
$$h = \frac{2(G\mathcal{M}_c)^{5/3}(\pi f)^{2/3}}{c^4 d}, \quad \frac{v}{c} = \left(\frac{GM\pi f}{c^3}\right)^{1/3}, \quad \dot{P} = -\frac{192}{5}\pi\left(\frac{G\mathcal{M}_c\pi f}{c^3}\right)^{5/3}$$

Source	GW150914	WD binary	SMBH (PTA)	H-T pulsar
$\mathcal{M} [M_\odot]$	30	0.3	10^9	1.23
d	400 Mpc	2 kpc	100 Mpc	6.4 kpc
f [Hz]	10^2	10^{-3}	10^{-8}	7.17×10^{-5}
h	9×10^{-22}	4×10^{-23}	3×10^{-15}	2×10^{-23}
v/c	0.5	2×10^{-3}	0.07	1.5×10^{-3}
$\dot{P}$ [s/s]	0.7	2×10^{-12}	5×10^{-5}	2×10^{-13}

Note the PTA column: $h \sim 10^{-15}$, six orders of magnitude larger than GW150914 — and still far harder to detect, because you have to wait years per cycle.

Two detectors, same *principle*.

LIGO — interferometer



Differential light-travel time between two perpendicular 4 km arms.

$$\Delta L \sim 10^{-18} \text{ m, over } \sim 0.1 \text{ s.}$$

PTA — precise clocks



Light-travel time from a millisecond pulsar to Earth.

$\sim 100 \text{ ns}$, over $\sim 15 \text{ years}$.

Both measure the **same quantity** — the fractional change in light-travel time between two points when a wave passes:

How loud is loud? — *SNR*.

You do not see the wave; you see it buried in noise with power spectral density $S(f)$. The optimal statistic is the matched filter, and its signal-to-noise is

$$\text{SNR}^2 = 4 \int \frac{|h(f)|^2}{S(f)} df = 4 \int \frac{h_c^2(f)}{f S(f)} \frac{df}{f}, \quad h_c \equiv f |h(f)|$$

Stationarity is doing the work: for stationary noise the covariance $\langle \tilde{n}(f) \tilde{n}^*(f') \rangle$ is **diagonal in frequency**, so each band is an independent measurement and SNR^2 is simply the sum of their contributions — that is the whole reason it is an integral over f . Writing it per logarithmic interval is what makes the plot two slides ago readable: the source curve sits above the detector curve exactly where the signal accumulates.

Monochromatic source

$$\text{SNR}^2 \sim \frac{|h_0|^2 f T}{f S(f)}$$

Grows as $\sqrt{T}$. This is the PTA and white-dwarf regime.

Chirping source

$$\frac{d \text{SNR}^2}{d \ln f} \sim \frac{|h_0|^2 f t_{\text{GW}}}{f S(f)}$$

The signal supplies its own integration time. This is LIGO.

The background is a dark, deep blue space filled with a fine grid of thin, light blue lines. Scattered throughout are numerous small, white, four-pointed stars. In the center, there is a large, swirling, elliptical structure resembling a galaxy or a nebula. This structure is composed of concentric, wavy bands of color, transitioning from dark blue and purple on the outer edges to bright yellow and orange in the center. The overall effect is one of cosmic depth and movement.

SECTION IV

A hundred years, and then ten

Some *history*.

1916 Einstein writes down GR; GWs predicted, then doubted.

1957 Chapel Hill: Feynman's sticky-bead argument settles that GWs carry energy.

1970S Interferometers and pulsar timing proposed.

1980S First ms pulsar; Hellings & Downs correlation.

1993 Nobel — Hulse & Taylor.

2017 Nobel — Weiss, Thorne & Barish. GW170817 multi-messenger.

2025 LVK passes 200+ merger events.

1936 Einstein submits "*Do Gravitational Waves Exist?*" Answer: *no*. Referee disagrees. Answer later flipped.

1963–64 Peters & Mathews compute the radiation from a Keplerian binary.

1974 Hulse–Taylor binary pulsar discovered.

1990 NSF funds LIGO. *Advanced LIGO comes online 2008*.

2015 GW150914 — first direct detection.

2023 PTAs announce evidence of a nHz background.

Note 1936 and 1957. Whether gravitational waves were even physical — as opposed to a coordinate artefact — was genuinely unsettled for decades.

The *Hulse–Taylor* pulsar.

A binary pulsar discovered in 1974, with

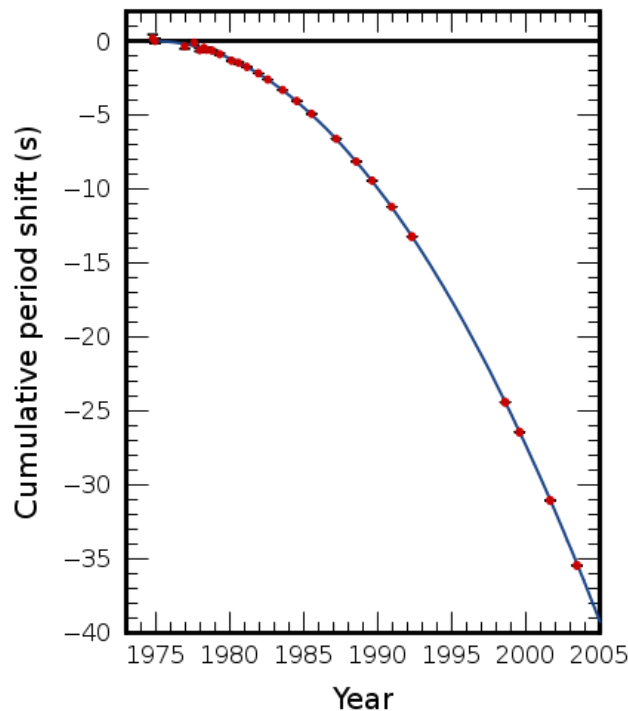
$$P_{\text{orb}} = 7.75 \text{ hr}, \quad e = 0.617, \quad t_{\text{merge}} \approx 300 \text{ Myr}$$

Separation about 1% of an AU — roughly the diameter of the Sun; $v \sim 10^{-3}c$.

Over decades it accumulated tens of thousands of orbits, decaying at exactly the rate predicted for energy loss to gravitational radiation. Agreement now better than 0.2%.

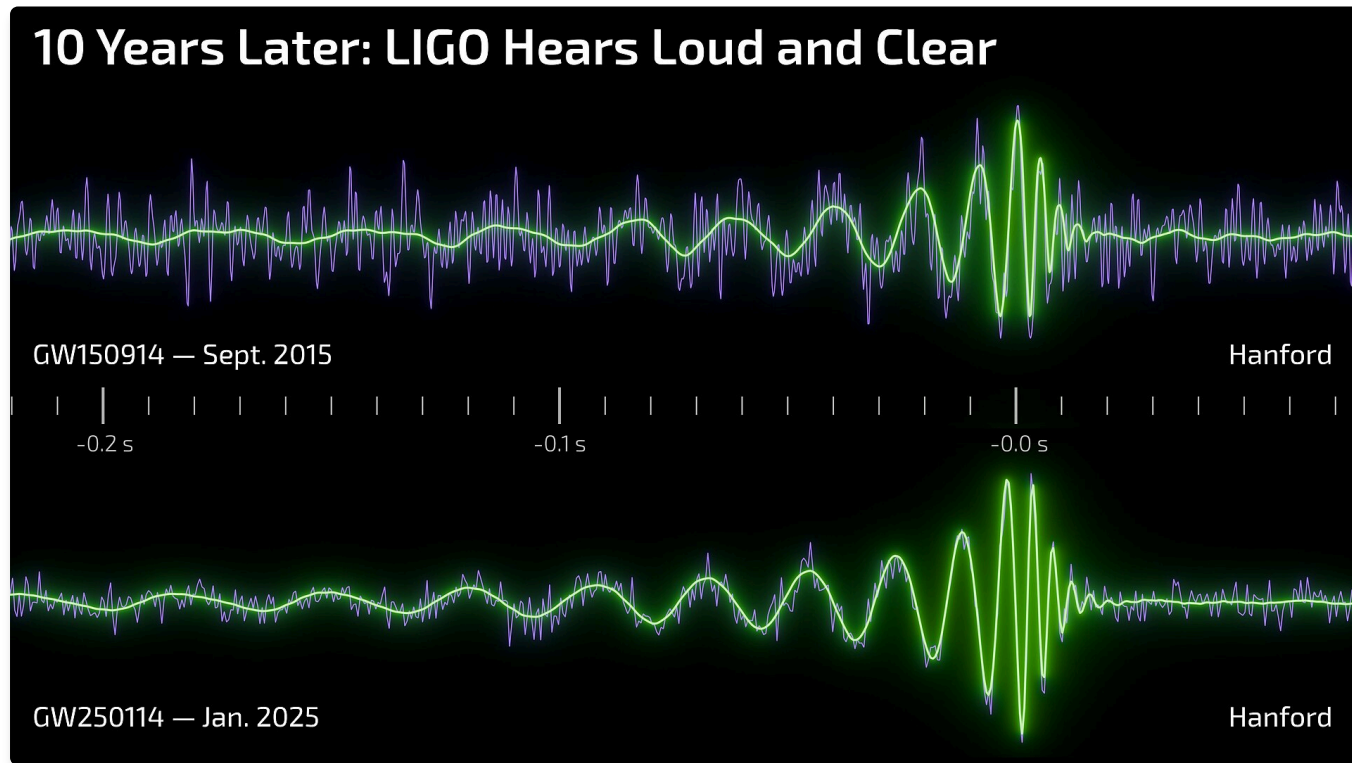
The first, indirect, proof — and a direct application of the $\dot{P}$ formula you derived earlier, with Peters' eccentricity correction.

NOBEL Hulse & Taylor, 1993



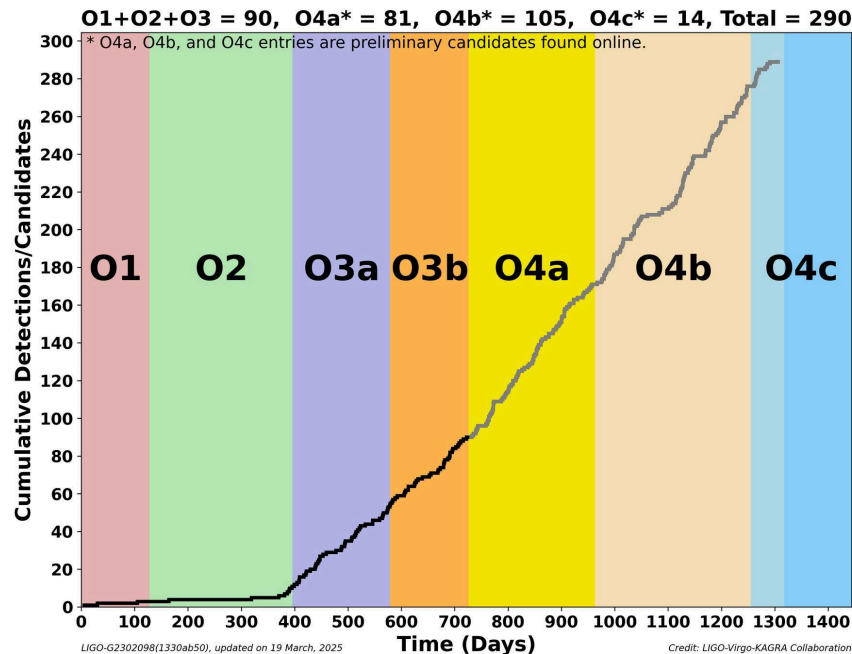
Cumulative period shift vs. year — data (points) vs. GR (line)

Ten years of *noise reduction*.



Same kind of event, ten years apart. The strain is the same; the noise around it has been beaten down. — LIGO Lab

A growing catalogue.



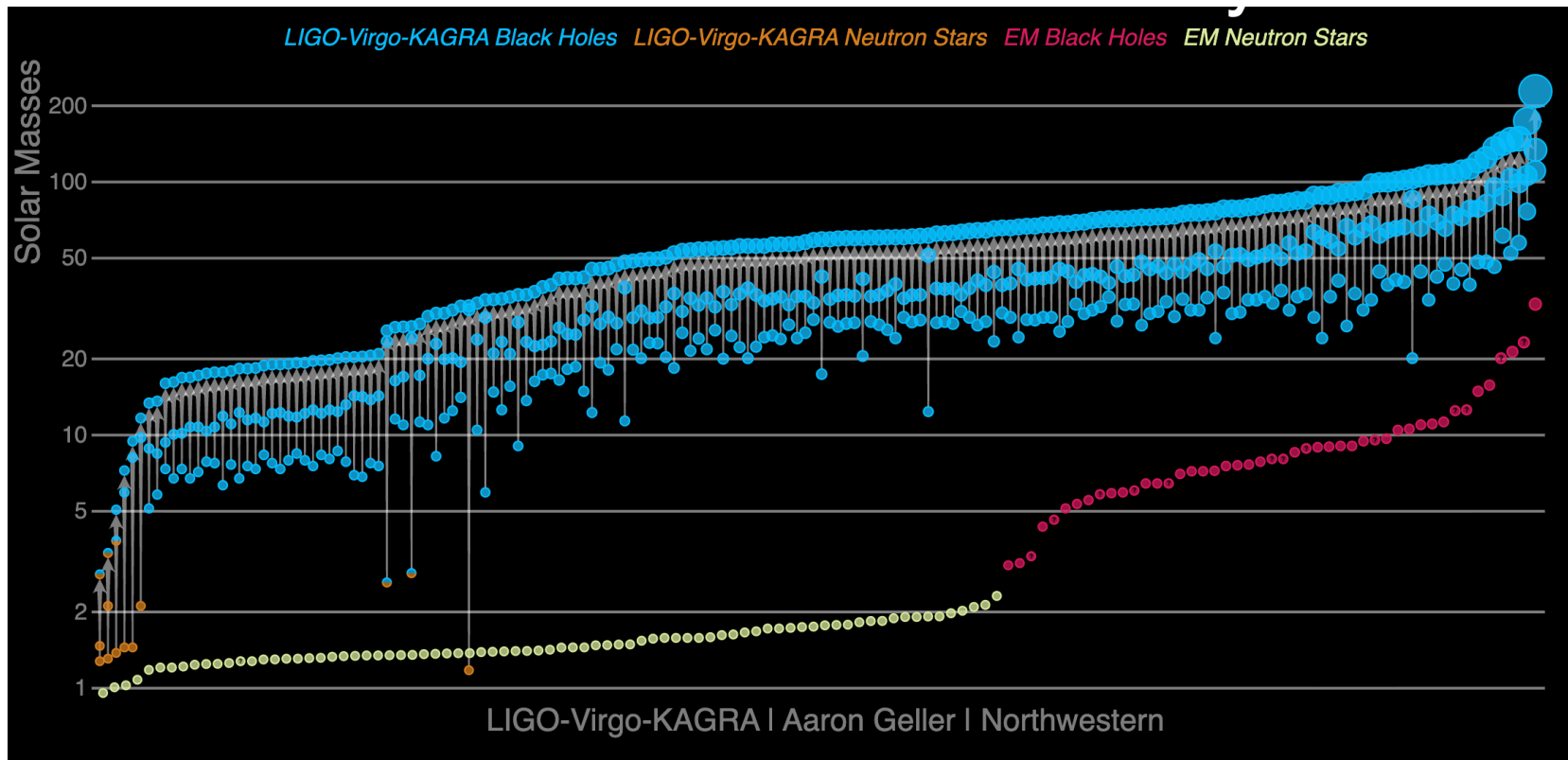
Cumulative LVK detections / candidates — LVK Collaboration, March 2025

By the end of O4 the catalogue is approaching **300 events**.

- The rate has gone up by an order of magnitude since O3.
- Each chirp lasts a fraction of a second to a few seconds.
- The strongest have SNR > 50.

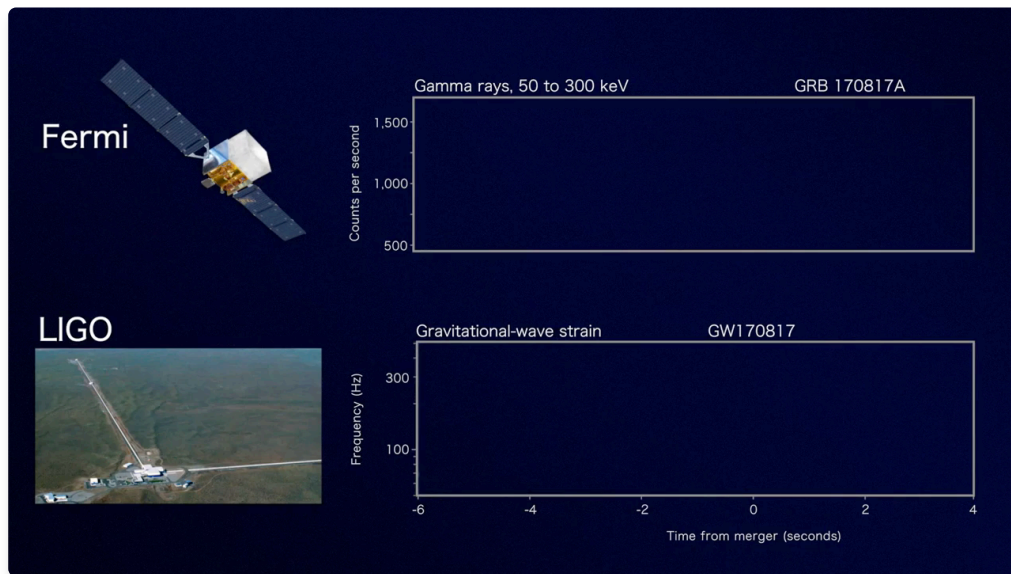
The waveform is computed from GR — post-Newtonian in the inspiral, numerical relativity through merger and ringdown.

Masses in the *Stellar Graveyard*.



Compact remnants observed via gravitational waves and through light. — LVK / A. Geller, Northwestern

GW170817 — a *multi-messenger* event.



Fermi gamma-rays & LIGO strain, GW170817 / GRB 170817A.

WHAT WAS SEEN

A 1.74-second delay between the gravitational-wave merger and the gamma-ray burst, over a 130-million-light-year baseline.

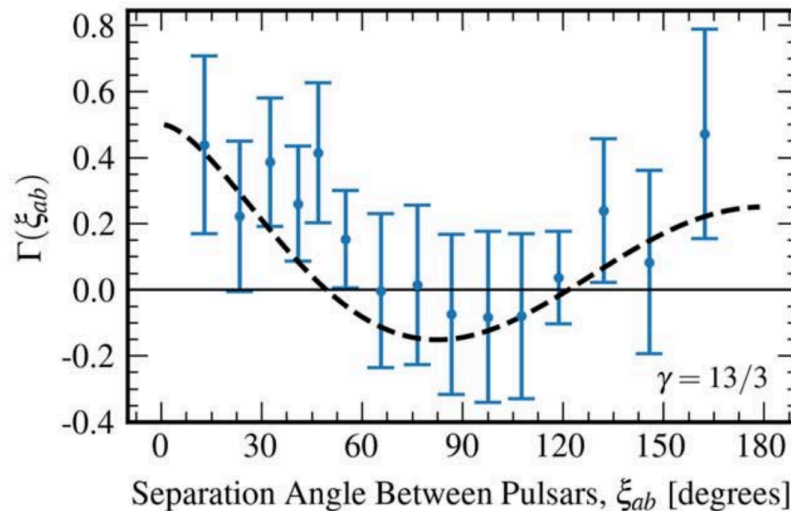
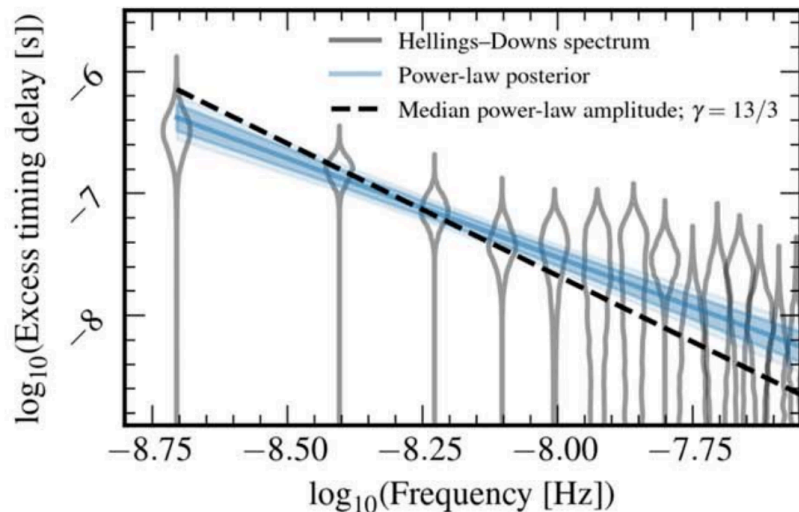
$$\frac{\Delta v}{c} \lesssim 3 \times 10^{-15}$$

WHAT IT BOUGHT

The speed of gravitational waves equals the speed of light to **a part in 10^{15}** — with no waveform calculation at all, just two arrival times.

Whole classes of modified-gravity theories died that day. Also: a kilonova, r-process nucleosynthesis, and an independent measurement of H_0 .

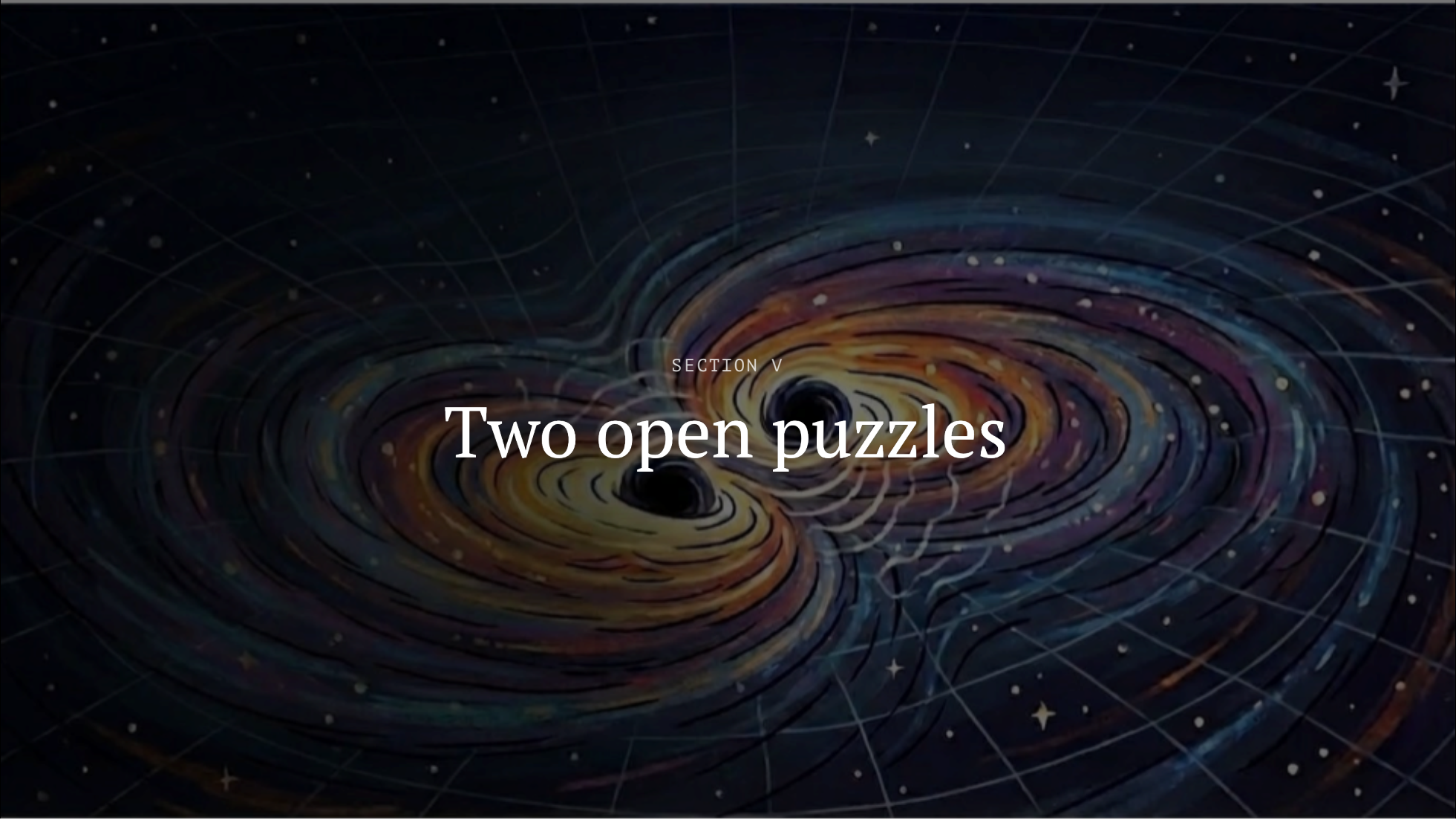
The nanohertz *hum*.



A passing wave shifts pulsar timing residuals in a **quadrupolar pattern** on the sky — spin 2 again.

The angular pattern is a parameter-free geometric prediction: the **Hellings-Downs curve**. You see it or you don't.

Now reported by **all major PTA collaborations independently** at $3\text{--}4\sigma$.

The background of the slide is a dark, deep blue space filled with a fine grid of thin, light blue lines. Two large, spiral galaxies are depicted in the process of merging. The galaxy on the left is primarily yellow and orange, with a bright, glowing core. The galaxy on the right is primarily blue and purple, also with a bright, glowing core. Both galaxies have prominent spiral arms. The text 'Two open puzzles' is centered over the merging region, and 'SECTION V' is positioned just above it.

SECTION V

Two open puzzles

Puzzle 1 — the black holes are *too heavy*.

Above $\sim 100 M_{\odot}$ a star's helium core gets hot enough that photons make e^+e^- pairs. Pressure support drops, the core contracts, oxygen burning runs away, and **the entire star is destroyed** — no remnant at all.

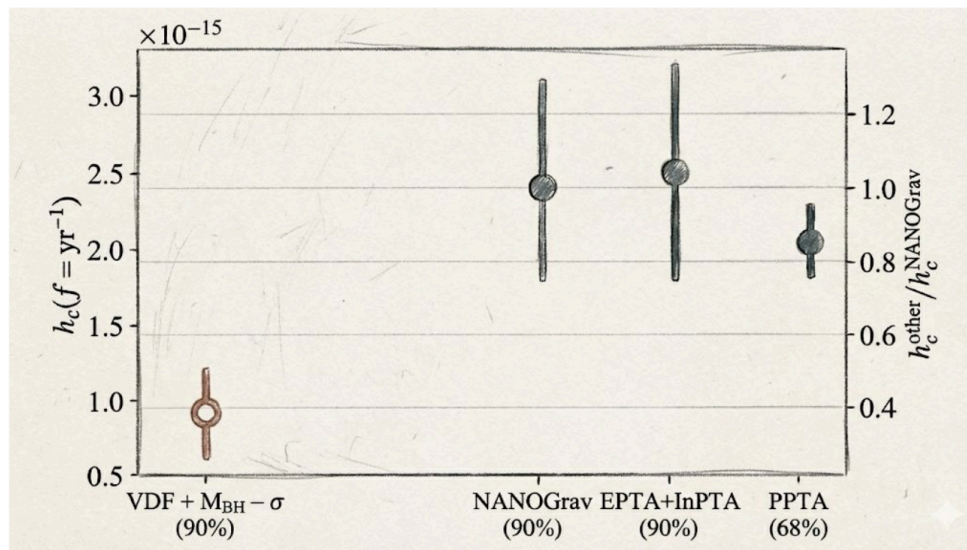
The prediction is a **forbidden zone** in the black-hole mass function, roughly $45 M_{\odot} \lesssim M \lesssim 130 M_{\odot}$.

Event	$m_1 [M_{\odot}]$	$m_2 [M_{\odot}]$	Notes
GW190521	~ 85	~ 66	first clearly-in-gap event
GW231123	~ 137	~ 101	most massive; both spins near extremal

And a second, quieter problem: to merge within a Hubble time, two $20 M_{\odot}$ black holes must orbit within $\sim 30 R_{\odot}$ — but their giant-phase progenitors were *hundreds* of solar radii across. How did they get so close?

You can check the first half of that sentence yourself — it is the Peters merger-time formula and nothing else.

Puzzle 2 — the nanohertz background is *too loud*.



Predicted vs. observed amplitude of the nHz background.

Phinney's argument — day 4 derives it — is that the background amplitude depends, to very good approximation, only on the **local mass density of supermassive black holes**:

$$\left. \frac{d\rho_{\text{GW}}}{d \ln f} \right|_0 = \rho_{\text{BH}} \langle \epsilon (1+z)^{-1/3} \rangle$$

Not on formation channel. Not on merger timescale.
Not on mass ratio.

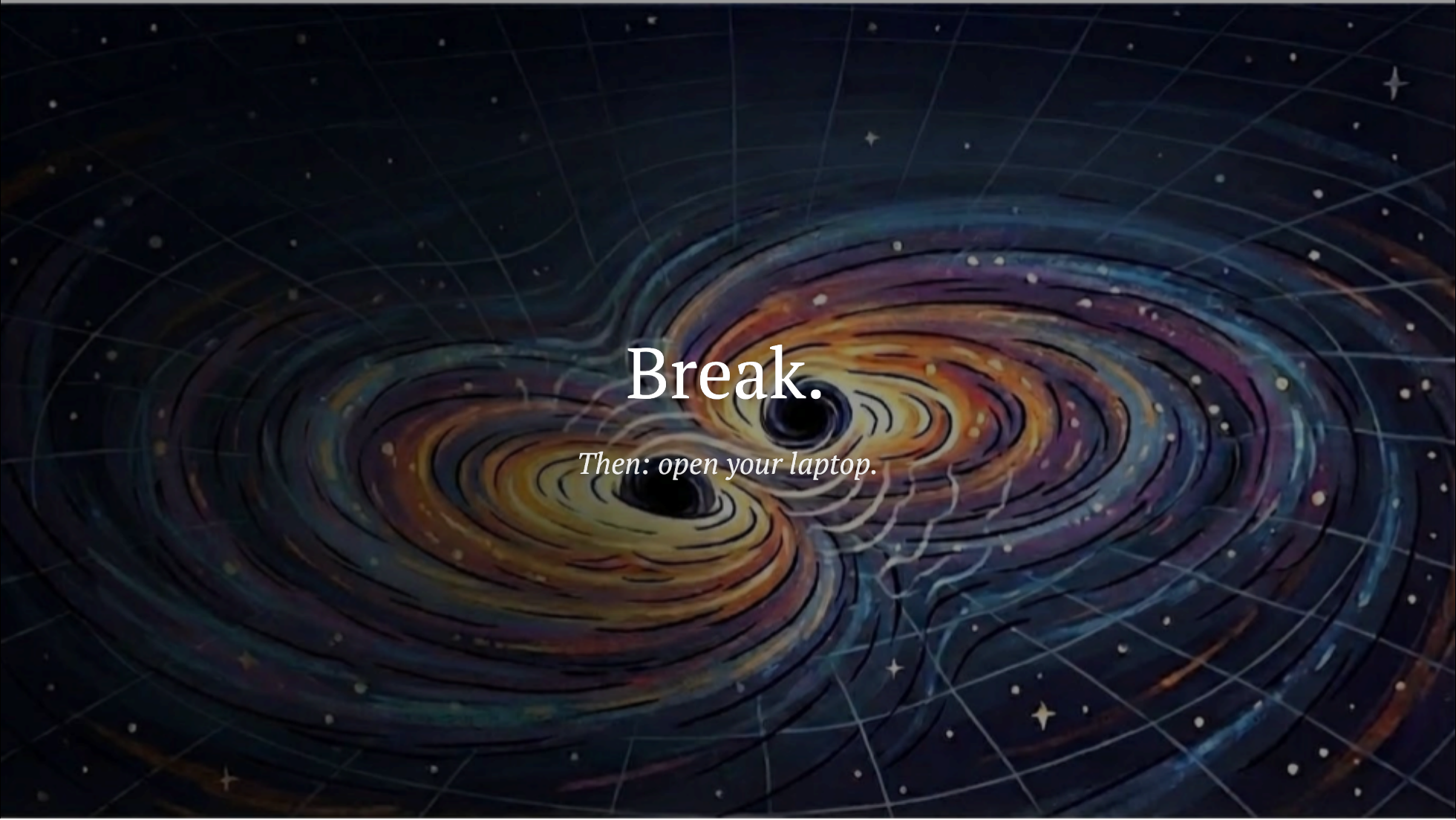
Fold in the local census from M- σ and the observed amplitude is **high by a factor of 2-3**.

Most natural reading: the local SMBH population has at least doubled its mass through mergers since accretion ended.

To take away.

- **Gravity is dynamical, so it has waves.** Two polarisations, spin 2, propagating at c , detectable only as a tidal effect between two free masses.
- **One formula, many masses.** $h \sim \mathcal{M}_c^{5/3} \Omega^{2/3} / d$ and $P \sim \eta^2 v^{10}$, with c^5/G as the only scale — an important formula, and almost every number today followed from it.
- **Mass sets frequency.** Ten decades of mass become ten decades of frequency; PTAs, LISA and LIGO are one subject.
- **We are ten years in,** and the objects we hear do not match the picture built from light.

Next: you will make an agent reproduce two of the calculations behind these statements — and we will find out how well that goes.

The background of the slide features a detailed illustration of two galaxies in the process of merging. The galaxies are depicted with vibrant, swirling arms in shades of blue, purple, orange, and yellow, set against a dark, star-filled cosmic background. A faint, white grid pattern is overlaid on the entire scene, suggesting a coordinate system or a digital space. The text 'Break.' is centered in a large, white, serif font, while the subtitle 'Then: open your laptop.' is positioned directly below it in a smaller, italicized, white serif font.

Break.

Then: open your laptop.